

Exp No: 25

PRODUCER–CONSUMER PROBLEM USING SYSTEMC

AIM:

To implement and simulate the Producer–Consumer problem using SystemC.

SOFTWARE REQUIRED:

- Ubuntu / WSL Linux
- g++ Compiler
- CMake
- SystemC Library

PROGRAM:

```
#include <systemc.h>
```

```
// Producer Module
```

```
SC_MODULE(Producer) {  
    sc_fifo_out<int> out;
```

```
    void produce() {  
        for (int i = 1; i <= 5; i++) {  
            cout << "Produced: " << i << endl;  
            out.write(i);  
            wait(1, SC_SEC);  
        }  
    }  
}
```

```
    SC_CTOR(Producer) {  
        SC_THREAD(produce);  
    }  
};
```

```
// Consumer Module
```

```
SC_MODULE(Consumer) {  
    sc_fifo_in<int> in;
```

```
    void consume() {  
        int value;
```

```

        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Consumer) {
        SC_THREAD(consume);
    }
};

int sc_main(int argc, char* argv[]) {
    sc_fifo<int> fifo(5);

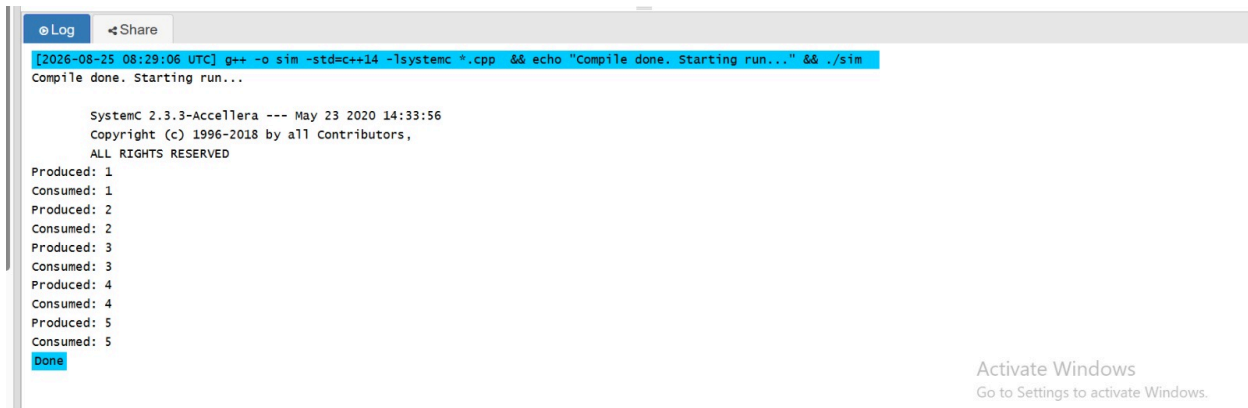
    Producer producer("Producer");
    Consumer consumer("Consumer");

    producer.out(fifo);
    consumer.in(fifo);

    sc_start(10, SC_SEC);

    return 0;
}

```



The screenshot shows a terminal window with the following content:

```

[2026-08-25 08:29:06 UTC] g++ -std=c++14 -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Done

```

At the bottom right of the terminal window, there is a watermark that says "Activate Windows Go to Settings to activate Windows."

RESULT:

Thus, the Producer–Consumer problem was successfully implemented and simulated using SystemC. Proper synchronization between concurrent processes was achieved using FIFO communication.